Exercise courses

Start with Exercises 1 and 2. Then choose an exercise 3 or 4 without assistance.
to 7 are a reserve of transfer, research or programming: the professor designates one.
Exercise 8 resumes at a distance.

Exercise 2.1 · Complete a square

The function is considered f defined on $\mathbb{R}$ by $f(x) = x^2 - 8x + 7$.

1. Complete the square to write $f(x)$ in form $(x - 4)^2 + k$ where: k is a real to be determined.
2. Deduce a factorization of $f(x)$ using a remarkable identity.
3. Draw up the signboard of f, then solve in $\mathbb{R}$ Inequation $f(x) \leq 0$.

Exercise 2.2 · Discuss without heavy calculation

Functions are studied $f(x) = x^2 + 4$, $g(x) = (x + 2)^2$ and $h(x) = -3(x - 1)(x + 4)$, all defined on $\mathbb{R}$.

1. For each function, determine any actual roots by directly exploiting the shape given.
2. Specify the sign of each function on $\mathbb{R}$. Indicate explicitly the values in which it cancels.
3. Explain why the calculation of a discriminant is not necessary in any of the three cases.

Exercise 2.3 · Selecting a Strategy

We wish to resolve in $\mathbb{R}$ the following three equations:

$$x^2 - 9 = 0, \qquad x^2 - 9x + 20 = 0, \qquad 2x^2 - 3x - 1 = 0.$$

1. For each equation, choose a method: remarkable identity, search for two numbers of sum and product known, or discriminating. Explain this choice.
2. Solve the three equations and give the solution sets in exact form.
3. Control the solutions of the first two equations by substitution. For the third, control the sum and product of the roots obtained.

Exercise 2.4 · Find a parabola

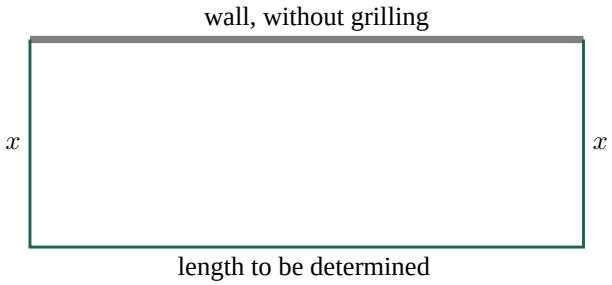
The curve of a polynomial function f of degree 2 cut the axis of the abscissas at the abscissa points -2 and 3. It also goes through the point $A(1; 12)$.

1. Determine an expression of $f(x)$ on the basis of this three information. Why she's okay.
2. Develop the term found. Deduce the order from the origin of the curve.
3. Verify that the function obtained satisfies the three initial informations.

Exercise 2.5 · A vegetable garden against a wall

A gardener has 20 m of grilling to surround three sides of a rectangular vegetable garden. fourth side runs along a wall and does not require grilling. x the length, in metres, of each of the two sides perpendicular to the wall. The whole length of the wire is used.

1. Make a diagram and express the length of the side parallel to the wall according to x. Specify the values of x for which the rectangle is not flattened.
2. To express the point $A(x)$, in square meters. Determine its maximum value and the corresponding dimensions, by choosing the method of your choice.
3. Check that these dimensions use exactly 20 m of grilling.



Vegetable garden diagram: only the three green lines use the 20 m of grilling. drawing does not fix the optimal dimensions.

Exercise 2.6 · A Parameter

For every real m, we consider the equation of actual unknown x :

$$x^2 - 2x + m = 0.$$

The number m is a parameter: it remains fixed when the equation is solved.

1. Determine the values of m for which the equation allows two distinct real solutions.
2. Determine the values of m for which it accepts a single real solution, then those for of which it does not admit any.
3. Give the solution in case it is unique. Explain why the three cases envisaged cover all values of m.

Exercise 2.7 · Link to the sequences

A sequence is defined by $u_n = n^2 - 5n + 4$ for all natural n. We're looking for all ranks to which the end of the sequence is strictly negative.

1. First solve in $\mathbb{R}$ Inequation $x^2 - 5x + 4 < 0$.
2. Deduce all natural integers n such that $u_n < 0$. Justify why a real interval does not This is not the final answer.

Fiscal year 2.8 · Deferred review

Student studying the function f defined on $\mathbb{R}$ by $f(x) = 2(x+3)^2 - 5$. She wrote: "The minimum value -5 and it is reached for $x = 3$. »

1. Indicate what is correct and what is wrong in this sentence. justifying your answer.
2. Solve in $\mathbb{R}$ Inequation $f(x) \leq -3$, then check the two bounds of the obtained set.

Personal work. Perform only the task assigned at the end of the meeting, in progress
Date: 10·20 min maximum. The exercises depend on the concepts already studied; some questions
Resuming exercise 8 a week after the chapter (10 min).
is exceeded, note the difficulty and bring the tests; do not accumulate the whole reserve.

To be done in the CQFD manual

These exercises are dealt with in your manual. The progress indicates the session; they replace the personal work planned that day (10 to 20 minutes in total).

CQFD, exercise 43 p. 99, questions a and c. First find an obvious root, then the other with the help of the root product.

CQFD, exercise 49 p. 99, questions a and b. Translate sum and product by equation ; do not assume in advance that the numbers exist.